

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	5
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	5
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	5
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	5
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	5
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	5

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	5
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	5
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	5
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	5

Code of Conduct:

I NITTHYENDREN JS / 192421180 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

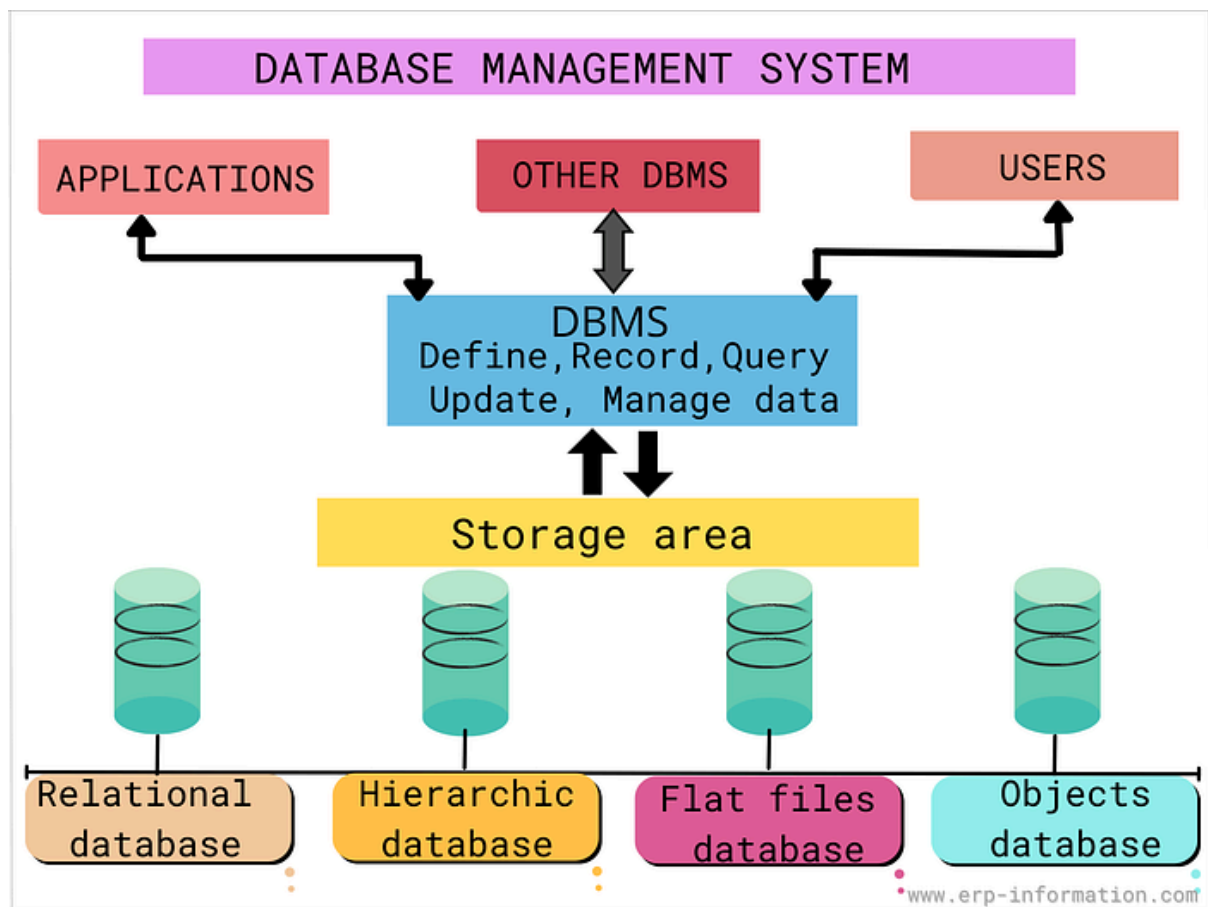
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
NITTHYENDREN JS

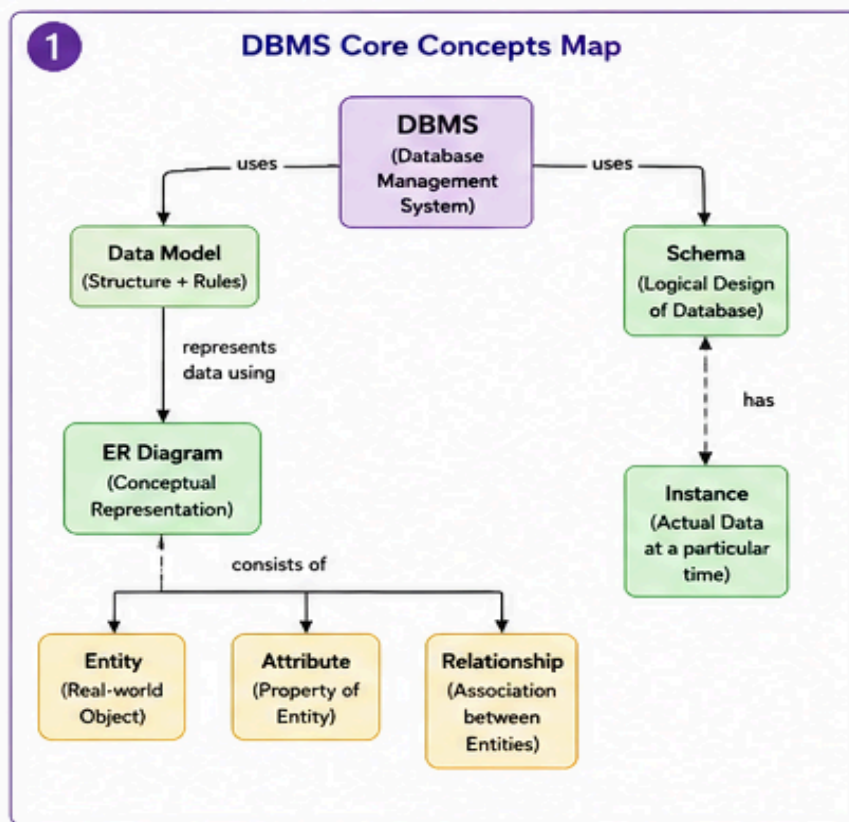
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

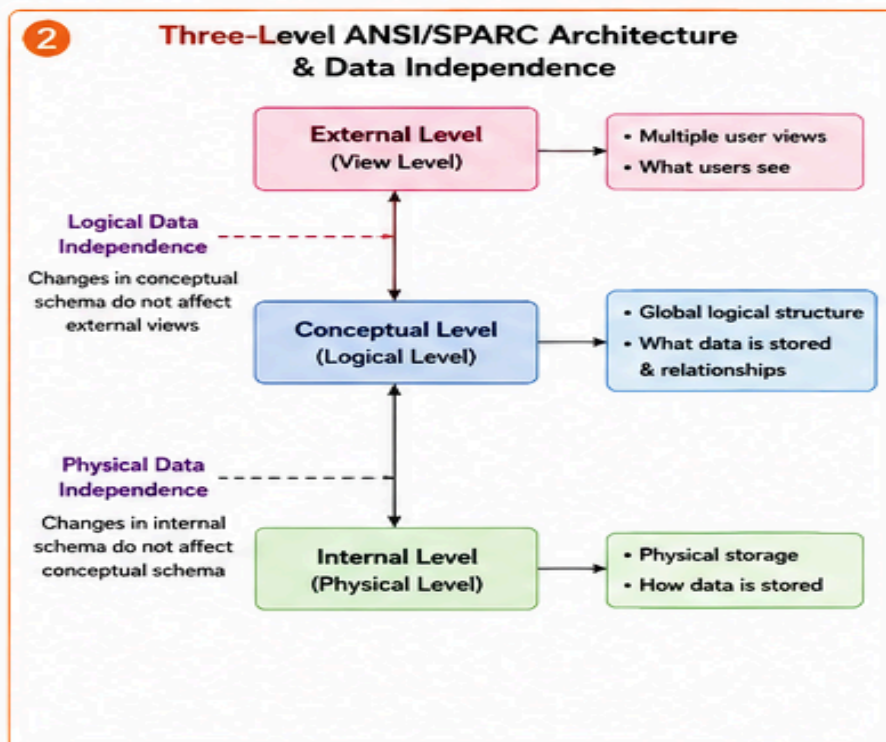
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand



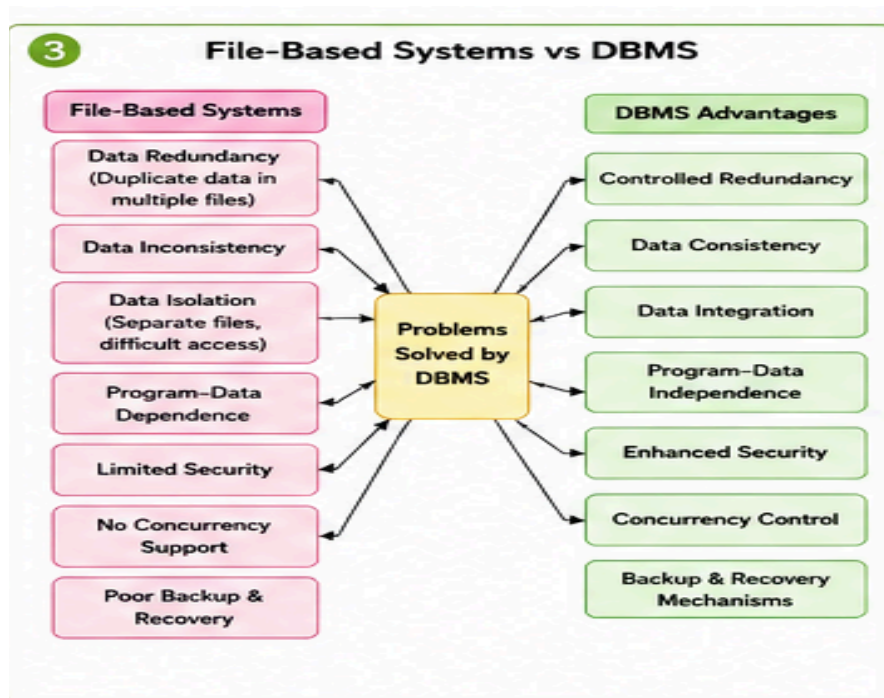
1.



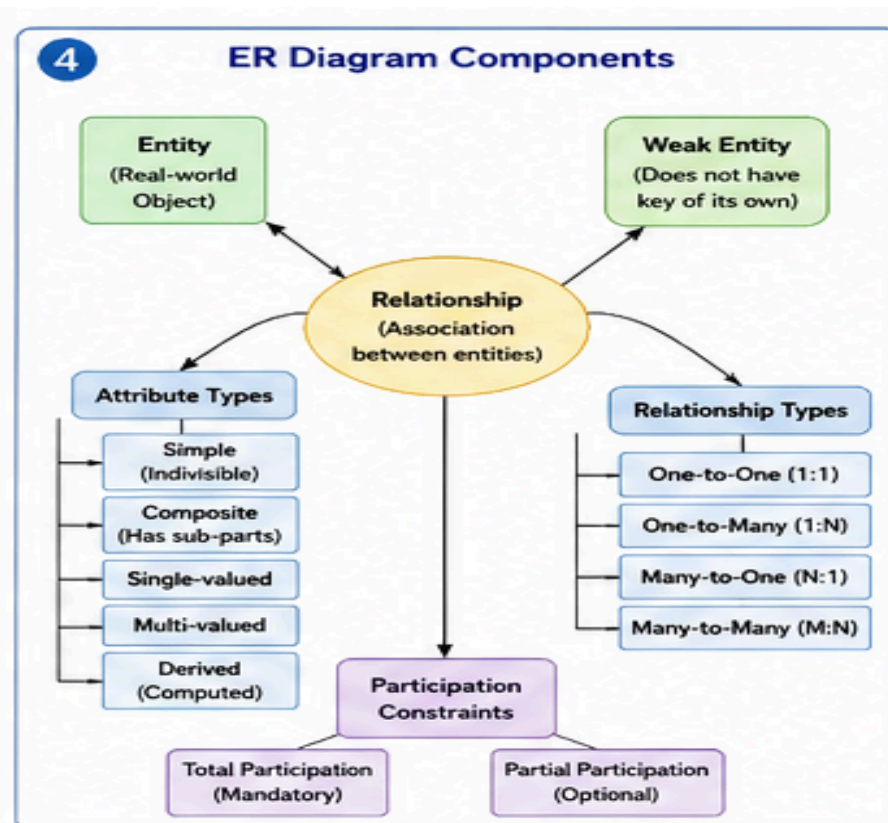
2.



3.



4.

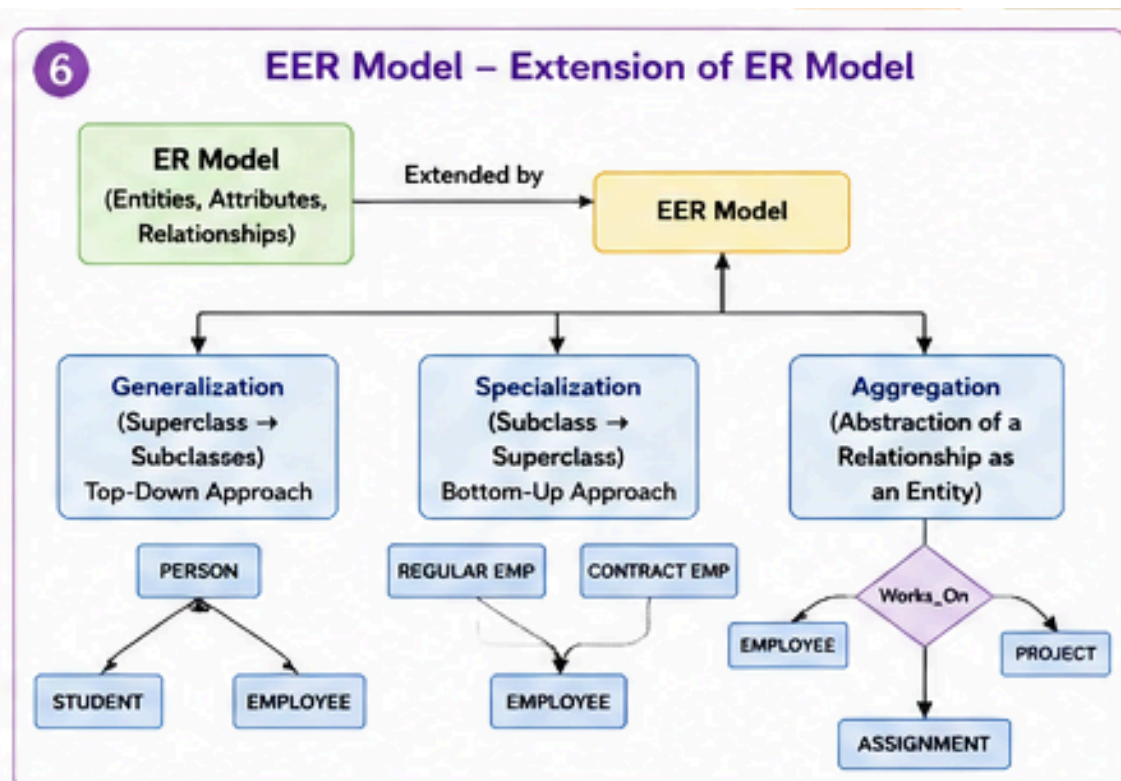


4.

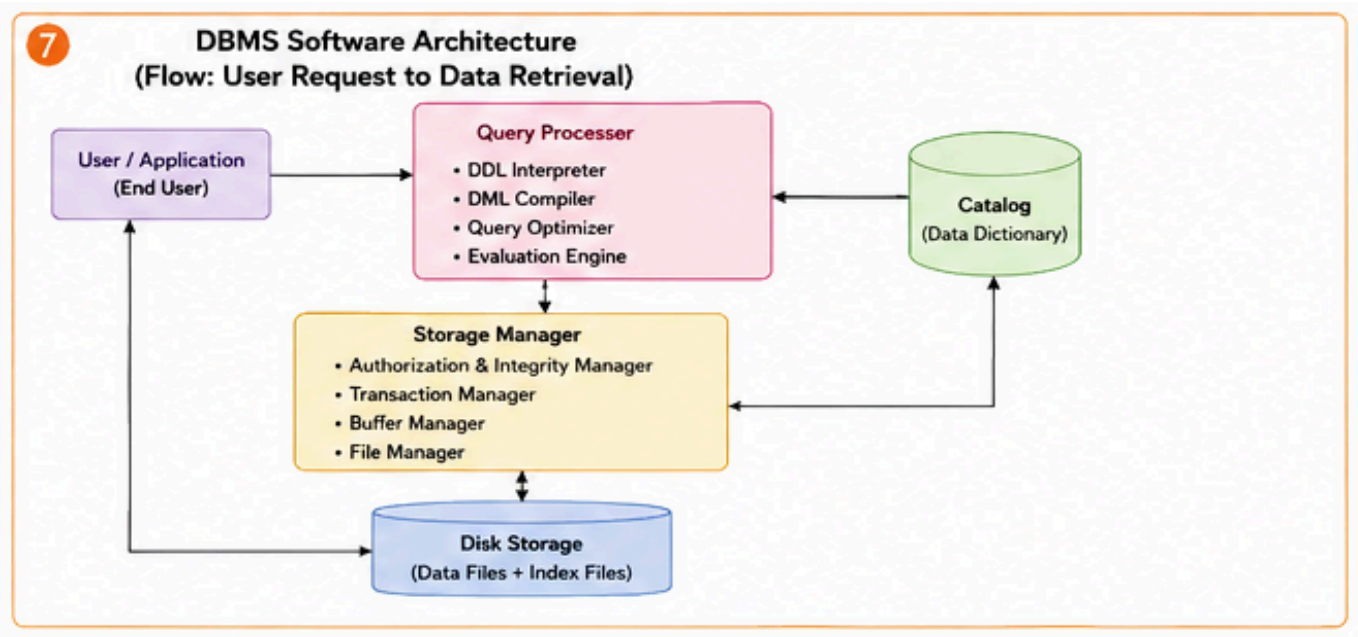
5.

5 Hierarchical, Network & Relational Data Models – Comparison			
Feature	Hierarchical Model	Network Model	Relational Model
Structure	Tree (Parent-Child)	Graph (Records linked in multiple ways)	Tables (Relations)
Relationship Type	One-to-Many	Many-to-Many	All via Keys (Primary & Foreign)
Data Representation	Records & Pointers	Records & Sets	Rows & Columns
Flexibility	Low	High	Very High
Data Independence	Low	Moderate	High
Ease of Use	Complex	Complex	Easy
Examples	IBM IMS	IDMS	MySQL, Oracle, PostgreSQL

6.



7.

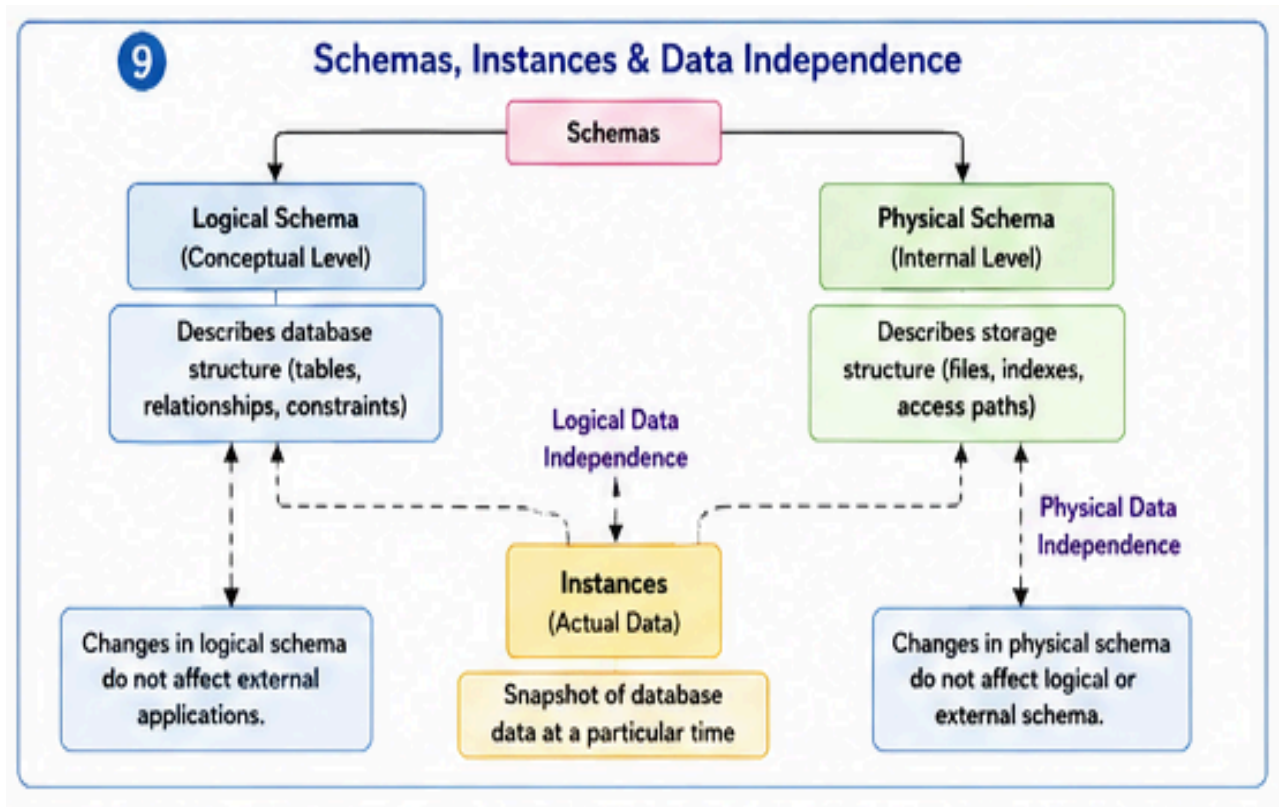


8.

8 Data Models & Real-World Applications

Data Model	Key Characteristics	Best Suited Applications	Why Suitable
Hierarchical Model	Tree structure, Parent-Child relationship	File systems, Org structures, XML data	Simple hierarchical data, fast access for one-to-many
Network Model	Graph structure, Many-to-many relationships	Telecom networks, Manufacturing, Supply chain	Handles complex relationships efficiently
Relational Model	Table (relation) based, uses keys (SQL)	Banking, ERP, CRM, E-commerce	Easy to use, flexible queries, high data independence
Object-Oriented Model	Objects, classes, inheritance	CAD/CAM, Multimedia systems, OOP apps	Handles complex data types & real-world objects
Object-Relational Model	Relational + Object features	Scientific apps, GIS, Data warehousing	Supports complex data + SQL capabilities

9.



10.

